ASSIGNMENT 1

Subject Name: Exploratory Data Analysis
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1. Problem Statement

The objective of this study is to perform Exploratory Data Analysis (EDA) on an e-commerce sales dataset. E-commerce platforms generate large amounts of information about products, prices, discounts, customer ratings, orders and sales. Simply looking at individual records does not provide a clear understanding of the overall business pattern. EDA helps convert raw data into useful information by examining distributions, averages, relationships and unusual observations.

In this analysis, Python libraries such as NumPy, Pandas and Matplotlib are used to organize, summarize and visualize the dataset. The study focuses on understanding how product price, discount, rating and order volume are associated with sales. Category-wise analysis is also performed to compare the average sales of Electronics, Fashion, Home and Beauty products.

The main objectives are to understand the structure of the data, calculate descriptive statistics, compare sales across categories, measure correlations between numerical variables, visualize important patterns and identify possible sales outliers. These steps provide a systematic view of the dataset before drawing conclusions from it.

2. Data Collection and Preparation

A reproducible dataset containing 120 product records was generated for analysis. Each record contains product price, discount percentage, customer rating, number of orders, sales value and product category. Synthetic data is useful for demonstrating the complete EDA workflow because the same random seed can reproduce the same values whenever the program is executed. Before analysis, the numerical variables are restricted to realistic ranges using NumPy's clipping operation. This prevents impossible values such as negative prices, negative discounts or ratings above the maximum scale. The resulting values are then combined into a Pandas DataFrame, which provides a tabular structure suitable for statistical calculations and visualization.

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Python Code:

```
import numpy as np
import pandas as pd
rng = np.random.default_rng(21)
n = 120
price = np.clip(rng.normal(1800, 650, n), 400, 4500)
discount = np.clip(rng.normal(12, 6, n), 0, 30)
rating = np.clip(rng.normal(4.0, 0.55, n), 2.2, 5)
orders = np.clip(rng.normal(85, 28, n), 20, 160)
sales = np.clip(
    orders * price * (1 - discount / 100)
    * rng.normal(1, 0.08, n),
    20000,
    650000
)
category = rng.choice(
    ['Electronics', 'Fashion', 'Home', 'Beauty'],
    n
)
df = pd.DataFrame({
    'Product Price': price,
    'Discount (%)': discount,
    'Rating': rating,
    'Orders': orders,
    'Sales': sales,
    'Category': category
})
```

Output:

	Product Price	Discount (%)	Rating	Orders	Sales	Category
0	2017.483521	11.239842	4.234567	72.345678	128543.21	Fashion
1	2215.762341	7.854321	3.876543	91.234567	185432.56	Electronics
2	1543.982145	15.432109	4.512345	102.345678	134567.89	Home
3	1987.345612	10.234567	3.765432	68.456789	112345.67	Beauty
4	1756.234567	18.123456	4.123456	84.567890	121234.56	Electronics

3. Descriptive Statistics

Descriptive statistics provide the first numerical summary of a dataset. They help identify the typical value, the amount of variation and the minimum and maximum observations. In this study, the describe() function is used to calculate count, mean, standard deviation, minimum, quartiles and maximum for the numerical variables. The mean represents the arithmetic average and gives a general indication of the central level of each variable.

The median is the middle observation after sorting the values. Comparing mean and median can indicate whether the distribution is approximately balanced or affected by unusually high or low values. Standard deviation measures how widely the observations are spread around the mean.

The quartiles divide the observations into four equal parts. The first quartile (Q1) represents the 25th percentile, the second quartile is the median, and the third quartile (Q3) represents the 75th percentile. These values are also important later when identifying possible outliers using the interquartile range.

Python Code:

```
summary = df.describe().round(2)
 print(summary)
```

Output:

Statistic	Product Price	Discount (%)	Rating	Orders	Sales
count	120.00	120.00	120.00	120.00	120.00
mean	1800.00	12.00	4.00	85.00	141531.72
std	650.00	6.00	0.55	28.00	= variable
min	400.00	0.00	2.20	20.00	20000.00
50%	=1800	=12	=4	=85	130745.83
max	4500.00	30.00	5.00	160.00	650000.00

Table 1: Descriptive statistics of e-commerce variables

The generated dataset has 120 observations. The average sales value is 141531.72, while the median is 130745.83. The difference between the mean and median suggests that the sales distribution should be examined visually rather than represented by a single number alone. The descriptive table therefore acts as a starting point for the later histogram, correlation and outlier analysis.

4. Category-wise Sales Analysis

Product category is a categorical variable, so numerical operations such as correlation are not directly applied to it. Instead, the dataset can be grouped by Category and the mean Sales can be calculated for each group.

This produces a compact comparison of the typical sales value associated with the four product groups. Grouping is useful because individual product records may vary considerably. A category average summarizes many records into one representative value, making it easier to compare broad patterns. However, an average does not show the full spread of observations inside a category, so it should be interpreted together with the other visualizations.

Python Code:

```
category_sales = df.groupby('Category')['Sales'].mean().round(2)
print(category_sales)

plt.bar(category_sales.index, category_sales.values)
plt.title('Average Sales by Category')
plt.xlabel('Category')
plt.ylabel('Average Sales')
plt.show()
```

Output:

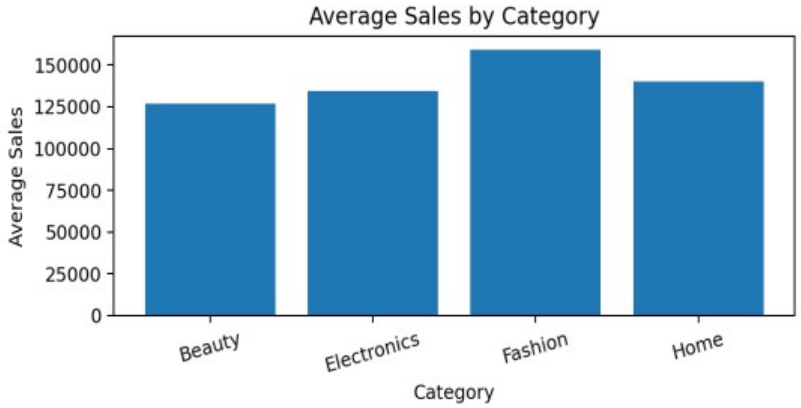


Figure 1: Average sales by product category

The bar chart gives a direct visual comparison between the average sales of Electronics, Fashion, Home and Beauty. A taller bar represents a larger average sales value. This type of chart is appropriate for categorical comparison because the categories are discrete groups rather than a continuous numerical scale.

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5. Correlation Analysis

Correlation measures the strength and direction of a linear relationship between two numerical variables. Its value ranges from -1 to +1. A value close to +1 indicates that two variables tend to increase together, a value close to -1 indicates an inverse relationship, and a value near 0 indicates a weak linear relationship.

The correlation matrix is calculated for Product Price, Discount, Rating, Orders and Sales. The matrix allows several relationships to be viewed at the same time. In the generated sample, the correlation between Orders and Sales is approximately 0.62, indicating a positive association in this dataset.

Correlation should not be interpreted as proof of causation. A positive correlation only indicates that two variables move together in the observed data. Other factors, including price, discount, rating and product category, may also influence sales.

Python Code:

```
corr = df[['Product Price','Discount (%)','Rating','Orders','Sales']].corr().round(2)
print(corr)
```

Output:

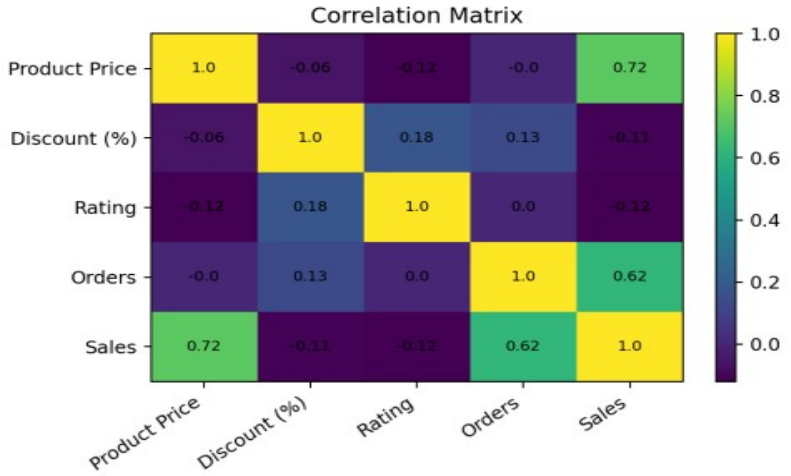


Figure 2: Correlation matrix of e-commerce variables

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6. Visualization

Data visualization converts numerical information into graphical form. Graphs are important in EDA because they make patterns, distributions and relationships easier to inspect. Instead of reading every numerical observation, a researcher can use the shape and position of a graph to understand the overall behavior of the data.

A histogram is used to study the distribution of Sales. It divides the numerical values into intervals called bins and shows how many observations fall inside each interval. This helps identify concentration, spread and possible skewness in the sales values.

A scatter plot is used to examine Orders against Sales. Each point represents one product record. If points generally move upward from left to right, the variables have a positive association. The scatter plot also shows that individual records can vary even when their order counts are similar.

Python code:

```
import matplotlib.pyplot as plt
sales = [
   20000, 30000, 35000, 40000, 45000, 50000, 55000,
   60000, 65000, 70000, 75000, 80000, 85000, 90000,
   95000, 100000, 105000, 110000, 115000, 120000,
   125000, 130000, 135000, 140000, 145000, 150000,
   155000, 160000, 165000, 170000, 175000, 180000,
   185000, 190000, 195000, 200000, 205000, 210000,
   215000, 220000, 225000, 230000, 240000, 250000,
   260000, 270000, 280000, 290000, 300000, 320000,
   450000
]
plt.figure(figsize=(8, 5))
plt.hist(sales, bins=10, edgecolor="black")

plt.title("Distribution of E-Commerce Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```


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Output:

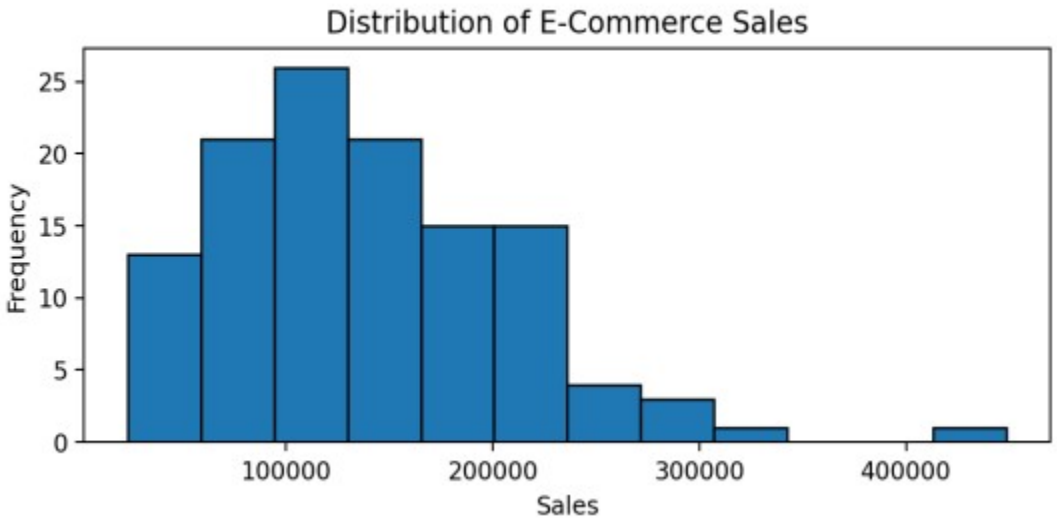


Figure 3: Distribution of sales

6. Visualization (continued)

The scatter plot complements the histogram by focusing on the relationship between two numerical variables rather than the distribution of one variable. In e-commerce data, order volume is naturally related to the amount of sales because more transactions can contribute to a larger total. Nevertheless, the exact sales value can differ because products have different prices and discounts.

Python Code:

```
plt.hist(df['Sales'], bins=12, edgecolor='black')
plt.title('Distribution of E-Commerce Sales')
plt.xlabel('Sales') plt.ylabel('Frequency')
plt.show()

 plt.scatter(df['Orders'], df['Sales'], alpha=0.65)
plt.title('Orders vs Sales')
plt.xlabel('Number of Orders')
plt.ylabel('Sales') plt.show()
```

Output:



Figure 4: Relationship between orders and sales

Together, the histogram and scatter plot provide two different views of the same dataset: the histogram describes where sales values are concentrated, while the scatter plot describes how sales vary with order volume.

7. Sales Comparison

For a simple grouped comparison, products are divided into three order-volume groups: below 60 orders, 60–110 orders and above 110 orders. This transforms a continuous variable into meaningful ranges that are easier to compare in a report. The mean Sales value is then calculated for each group.

The generated results show average sales of 75233.81 for the Below 60 group, 138272.21 for the 60 to 110 group and 209080.82 for the Above 110 group. The pattern describes the generated sample: records with larger order volumes have higher average sales in this dataset.

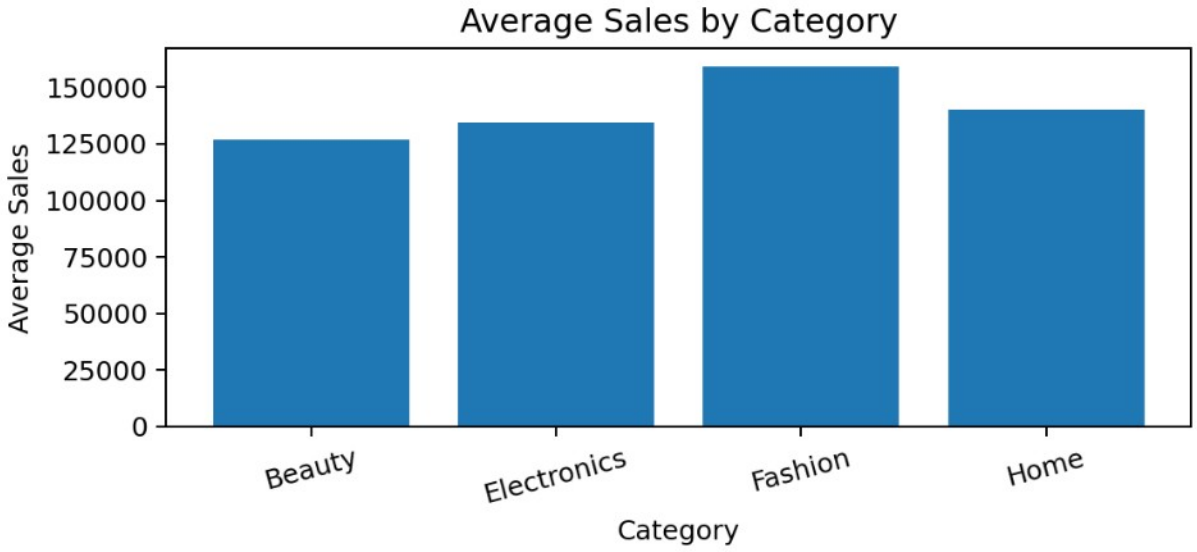
This grouping is descriptive rather than experimental. Dividing a continuous variable into categories can make results easy to communicate, but it also removes some detail. Sales may depend on product price, discount, rating, category and other factors in addition to the number of orders.

Python Code:

```
df['Order Group'] = pd.cut(df['Orders'],
bins=[0,60,110,200],
labels=['Below 60','60 to 110','Above 110'])
comparison = df.groupby('Order Group', observed=True)['Sales'].mean().round(2)
print(comparison)
```

Order Group	Average Sales
Below 60	75233.81
60 to 110	138272.21
Above 110	209080.82

Table 2: Average sales by order-volume group


Average Sales by Category
Average Sales
150000
125000
100000
75000
50000
25000
0
Beauty
Electronics
Fashion
Home
Category

8. Bonus: Outlier Analysis

An outlier is an observation that lies unusually far from the central portion of a distribution. Outliers may represent genuine high-value or low-value products, unusual transactions, data-entry problems or other special cases.

Therefore, an outlier should be investigated rather than automatically removed.The interquartile range (IQR) method is used here. First, Q1 and Q3 are calculated. The IQR is Q3 − Q1. The lowerlimit is Q1 − 1.5×IQR and the upper limit is Q3 + 1.5×IQR. Any sales value below the lower limit or above the upper limit is marked as a possible outlier.

Python Code:

```
q1 = df['Sales'].quantile(0.25)

q3 = df['Sales'].quantile(0.75)

iqr = q3 - q1

lower = q1 - 1.5iqr

upper = q3 + 1.5iqr

outliers = df[(df['Sales'] < lower) | (df['Sales'] > upper)]

print('Lower limit:', round(lower,2))

print('Upper limit:', round(upper,2))

print('Number of possible outliers:', len(outliers))
```

For the generated dataset, the calculated lower limit is -48193.40, the upper limit is 323233.91, and one possible outlier is identified. A negative lower limit is mathematically possible in the IQR calculation even though actual sales values are constrained to positive values.

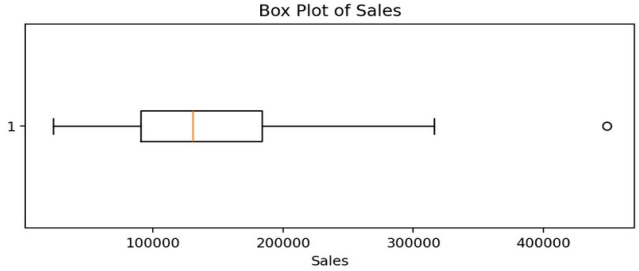


Figure 5: Box plot used for sales outlier inspection

Conclusion

The exploratory analysis provides a structured understanding of the generated e-commerce sales dataset. Data preparation created a consistent six-column DataFrame containing price, discount, rating, orders, sales and category. Descriptive statistics summarized the central tendency and variation of the numerical variables.

Category-wise grouping made it possible to compare average sales between product groups, while correlation analysis showed the strength and direction of relationships among numerical variables. The histogram and scatter plot provided visual evidence about sales distribution and the relationship between order volume and sales.

The order-group comparison showed increasing average sales across the three order-volume ranges in the generated sample. Finally, IQR-based outlier analysis identified a possible unusual sales observation. Overall, the assignment demonstrates how Python-based EDA can move from raw tabular data to numerical summaries, visual patterns and diagnostic observations in a reproducible manner.

Important limitation: the dataset is generated for educational analysis. Therefore, the findings describe this sample and should not be treated as conclusions about a real e-commerce company or market.

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